

Tracking the Evolution of AI, Machine Learning, and NLP Research Discourse (2000–2021) from Theory to Application

Group Report

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1 Abstract

The development of scientific discussion is both a practical question about a research area and a test of text analysis methods. This study examines how AI, machine learning, and NLP research summaries shifted from focusing on theory to practical use, how they altered how they describe limitations, and how the spread of new terms accelerated over the last thirty years. To study these changes, five methods were applied to 24,677 arXiv summaries: keyword analysis, topic modeling (LDA and NMF), classification of how limitations are described, tracking how terms spread (TF-IDF changes, Word2Vec, and SPECTER2), and dimension reduction (PCA and UMAP). Each method provides independent evidence, and all point to 2014–2016 as the main turning point. For example, TF-IDF and Sentence-BERT (all-MiniLM-L6-v2) centroid distance series have a Pearson correlation of $r = 0.863$. Important results show that language focused on applications passed theory-focused language around 2014–2015; ethics-related terms grew from almost none to about 13% of summaries by 2020–2024; and UMAP suggests the field shifted to new approaches rather than replacing old ones. Also, variation within each year drops significantly (Kendall's $\tau \approx -0.5$, showing the field is becoming more consistent as it develops. Most importantly, the main way terms change is by quietly shifting meaning rather than by being used more often: 13 of 36 tracked terms changed meaning without a matching increase in use, a result that frequency-based methods alone would miss.

2 Introduction

2.1 Motivation

Research language is not neutral. Abstracts show what a field thinks is important, what counts as a valid contribution, and what claims need support. As AI, machine learning, and NLP have grown from small academic areas into technologies that affect daily life, the language used to describe this research has changed too, but how to measure that change and which tools best capture it remain unclear.

Earlier research in computational scientometrics has examined citation networks, co-occurring keywords, and topic patterns to track changes in the field [1, 2]. These studies usually use a single way of representing data and a single method of comparison, making it hard to distinguish real changes in the discussion from errors introduced by the method. For example, a rise in how often certain words are used could indicate a genuine shift in ideas, a trend in language, or just a shift in where papers are published. Results that appear across different methods with very different assumptions are much more reliable.

2.2 Research Questions

This study examines three interconnected research questions. First, how have AI, ML, and NLP research summaries shifted their focus from basic theory to real-world uses between 1990 and 2025? Second, have the ways limitations are described in summaries changed, and has the growth of AI systems in use brought new ethical concerns? Third, how does specific technical language spread in a research field, mainly by how often it is used, or by changing meaning in new discussions? A fourth overall question drives the mixed-method approach: do different text analysis methods agree on when and how these changes happened, or do they find very different patterns?

2.3 Corpus and Motivating Observation

The study is based on 24,677 arXiv abstracts, mainly from the cs.LG, cs.CL and cs.AI, covering the years 2000 to 2021. The number of publications is highly uneven, ranging from 261 abstracts from 2000 to 2004 to 7,677 in 2020 to 2021. The focus of the categories changes from mostly cs.CL in the early years to mostly cs.LG after 2015. This means the collection is not a uniform sample of AI research over time but a changing mix of different subfields, which any analysis must clearly consider.

The extent of language change can be clearly shown. A typical abstract from 2003 says: “*We present a Bayesian model for word sense disambiguation that infers posterior sense distributions from prior lexical knowledge and contextual co-occurrence statistics.*” These two texts use almost no common words and make very different knowledge claims. The first explains a theoretical model, while the second shows test results. Carefully measuring this change across 24,677 documents is the main challenge of this paper.

2.4 Why a Multi-Method Design

No single text analysis method can address all aspects of this problem simultaneously. Keyword analysis is easy to understand, but it only works with a fixed set of words. Topic modeling finds hidden themes without requiring predefined words, but it relies on simple word counts and requires manual labeling. Dimensionality reduction works regardless of data type but merges differences within the same year and is affected by changes in subfield makeup. Distributional drift analysis separates how often words are used from changes in their meaning, but it requires careful matching of time periods. Supervised classification measures changes over time directly, but it confuses topic words with the specific signals it aims to detect. Importantly, these weaknesses balance out because each method’s strengths compensate for others’ limitations. This is why we use five methods here. A result that appears independently across several methods is much more reliable than one found in a single analysis.

3 Methodology

3.1 Dataset and Target Scope

The dataset is made from gfsiore/arxiv-abstracts-2021 on HuggingFace Hub (1,999,486 preprints)[3]. To build it, only abstracts where the main category is the first author-assigned field in cs.LG, cs.CL, cs.AI are kept. Papers listed with these categories only as secondary are excluded because their main argument is based on their primary subfield. After requiring abstracts to have at least 50 words and to include both year and category information, 24,677 abstracts remain. For time-based analysis, five-year periods (2000–04 to 2020–21) are used. This is because yearly groups were too small before 2005 (only a few hundred), while ten-year groups would mask the 2014–2015 change observed in the baseline.

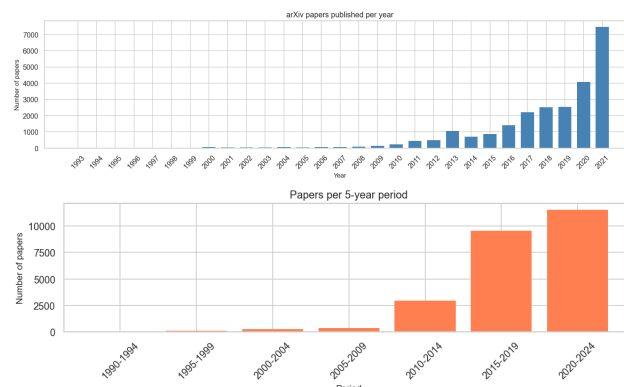
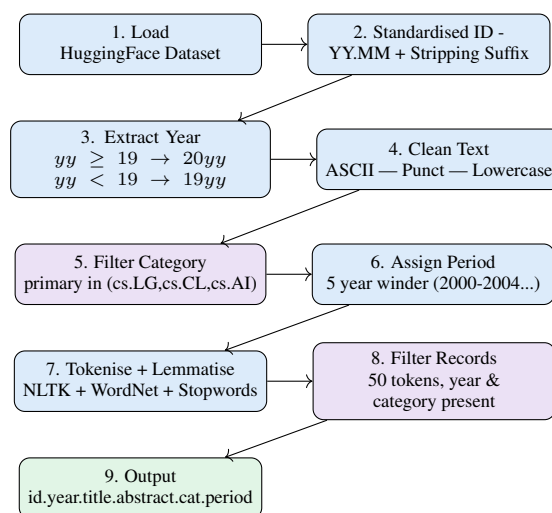


Figure 1: Publication volume over time, showing the 27-fold skew from 2000–2004 to 2020–2021 that motivates period-level reliability weighting.

3.2 Preprocessing Pipeline



The preprocessing pipeline is designed to support both traditional bag-of-words models and transformer-based embeddings. For bag-of-words methods, we apply aggressive abstract cleaning to reduce noise and standardise lexical features, whereas raw abstracts are preserved separately for SentenceBERT and SPECTER2 [4] to retain semantic richness. Text normalisation includes ASCII filtering, punctuation removal, and lowercasing, followed by tokenisation and WordNet-based lemmatisation to reduce vocabulary sparsity.

A domain-specific stopword list of 24 terms (e.g., *paper, study, work, approach, used, based*) is applied in addition to the standard NLTK English stopwords to further eliminate high-frequency academic boilerplate. Finally, records are filtered for minimum content length and completeness, and enriched with structured metadata including temporal periods and category labels for downstream analysis.

3.3 Text Representations

Five representations are used across the project; each is chosen to make a different layer of change visible: surface vocabulary, distributional neighbourhood, or full discourse

Table 1: Representation models and semantic interpretation layers

Representation	Level	Interpretation / Use
BoW (term counts)	(term raw frequency)	LDA baseline; keyword-frequency model used as a lexical reference
TF-IDF (sublinear)	discriminative vocabulary	Keyword baseline used in NMF; sensitive to vocabulary choice and PCA-related instability
Word2Vec (skip-gram)	local neighbourhood semantics	Captures terminology diffusion via co-occurrence drift in embedding space
Sentence-BERT (MiniLM-L6)	abstract semantic representation	Centroid drift analysis using PCA and UMAP projection of semantic structure
SPECTER2	scientific discourse embedding	Models terminology diffusion and discourse-level semantic drift across scientific literature

3.4 Lexical Keyword Analysis (M5)

The baseline operationalises theoretical versus applied focus as the relative prevalence of two hand-curated keyword sets. An initial vocabulary of 14 theory and 15 application terms was reduced after exploration: high-frequency ambiguous terms such as *model, method, results, and framework* appeared with nearly equal frequency in both orientations and flattened the signal. The retained sets theory: {*theory, hypothesis, framework, approach, theorem*}; application: {*application, dataset, experiment, implement*} trade coverage for signal clarity. Two scores are computed per year y and keyword set K :

$$\text{RTF}(y, K) = \frac{\sum_{d \in D_y} \sum_{k \in K} \text{count}(k, d)}{\sum_{d \in D_y} |d|_{\text{chars}}} \quad (1)$$

$$\text{tfidf}(t, d) = (1 + \log \text{tf}(t, d)) \cdot \log \frac{|D|}{|\{d' : t \in d'\}|} \quad (2)$$

Character-level normalisation in Eq. 1 was chosen over per-document averaging because post-filter abstract lengths span 50–400 tokens. Sublinear TF in Eq. 2 is essential; without it, near-function terms dominate year-level means. Both representations place the crossover at 2014–2015.

3.5 Topic Modelling

Topic modelling removes the closed-vocabulary constraint of the baseline and allows themes to emerge from the data. LDA (Gensim) on a bag-of-words matrix and NMF (scikit-learn) on a TF-IDF matrix were trained and compared for $k \in \{6, 8, 10\}$ topics. Model quality is evaluated with C_v coherence, which tracks co-occurrence of top-weighted words within a sliding reference window and is the standard human interpretability proxy for short-document corpora. Held-out perplexity was deliberately excluded as it rewards diffuse topics and correlates poorly with human judgments on short documents. NMF with $k = 10$ was selected ($C_v = 0.55$ vs. 0.48 for LDA at the same k); topics were manually labelled as *theory-oriented* (algorithm, optimisation, inference) or *application-oriented* (dataset, healthcare, translation, recommendation) and aggregated into a time-wise theory/application ratio directly comparable with the lexical baseline.

3.6 Limitations Framing and Classification

To check if the type of limitation mentioned in abstracts has changed, keywords were divided into four separate groups: *failure* (fail, degrade), *capability* (cannot, limited), *hedging* (may, could), and *ethics* (bias, fairness). For each abstract, a yes/no marker shows if any keyword appears in each group, and the average proportion is calculated across groups. Another test uses logistic regression on TF-IDF word features to tell apart 2010–2013 from 2018–2020 abstracts, two separate time periods chosen to avoid confusion from transition years. Five-fold cross-validation is applied; per-class F1 scores are reported along with accuracy because there are more late-period abstracts than early ones, and the confusion matrix is checked for uneven errors, where late abstracts are wrongly labeled as early ones to find leftover early-period words in late abstracts.

3.7 Representation Geometry and Temporal Drift

To test whether the shifts identified lexically and thematically leave a geometric trace, each abstract is embedded, and per-year centroids are compared. TF-IDF captures surface vocabulary change; SentenceBERT (all-MiniLM-L6-v2, mean-pooled) captures semantic change. For visualisation, PCA (2D) and UMAP are both applied: PCA preserves global variance while UMAP preserves local neighbourhoods; both are reported because a single projection can mislead. Quantitative drift is computed in full-dimensional space—50-component PCA for TF-IDF and 384-dimensional space for SBERT—because the 2D TF-IDF projection retains only approximately 1.6% of total variance. Temporal movement is the consecutive year centroid distance:

$$d(y) = \|\mu(y) - \mu(y-1)\|, \quad \mu(y) = \frac{1}{|D_y|} \sum_{d \in D_y} \phi(d) \quad (3)$$

Series are $[0, 1]$ -normalised for cross-representation compar-

ison. Robustness is confirmed by near-perfect agreement between Euclidean and cosine distance ($r \approx 1.00$), confirming observed patterns are properties of the representation, not the metric.

3.8 Terminology Diffusion (M2)

To follow individual terms through the big change, three matching signals are calculated for each candidate term in each time period. Candidate terms are identified by retaining the top 60 rising TF-IDF terms for each period; about 37% are removed because they are part of longer phrases, leaving around 48 candidates. Before training Word2Vec, 174 multi-word phrases are combined into single tokens with underscores to avoid incorrect splitting.

- (i) **TF-IDF delta** delta weighted late minus early period mean frequency per 10,000 words measures surface adoption.
- (ii) **Word2Vec neighbourhood cosine**, calculated between skip-gram models made for each period (vector size 300, window 12, epochs 30), measures how much a term's context changes; the models are aligned using a Procrustes orthogonal transformation based on 1,436 shared anchor terms (6.9% of the combined vocabulary), with reliability scores of 0.27 and 0.49 for the two earliest periods.
- (iii) **SPECTER2 discourse drift** (allenai/specter2, HuggingFace) captures argumentative repositioning at the abstract level; embedding operates on raw non-lemmatized abstracts (batch size 64, L2-normalized) with bootstrap aggregation stabilizing small-sample cells.

All three signals are scaled between 0 and 1 and cut off at 0.5 to create a four-part classification that separates terms spreading actively from those copied from templates, where high frequency and almost no change in discourse indicate uniform terminology rather than real idea spread.

4 Analytical Framework and Hypotheses

The analysis is organized around two main parts: how the text is represented and how changes are compared. Together, these determine which types of changes can be observed and how reliably they are detected. Using different ways to represent and compare the same set of texts helps assess reliability by checking whether different methods agree. When results are supported by multiple methods, they are more trustworthy than those from a single method.

Axis 1 – Text Representation

This axis concerns how abstracts are encoded. Lexical representations (term frequency, TF-IDF) capture surface vocabulary change and provide interpretability, but cannot detect semantic shift when wording remains stable. Topic models (LDA, NMF) infer latent thematic structure from co-occurrence patterns, enabling detection of broader shifts

in research focus, but introduce modelling assumptions and require manual interpretation.

Embedding-based representations (Word2Vec, Sentence-BERT, SPECTER2) capture contextual and semantic similarity, allowing detection of conceptual and discourse-level change even when vocabulary is stable, but are less interpretable and require triangulation. Together, these representations separate lexical change (what is said), thematic change (what is discussed), and semantic change (how ideas are framed).

This axis improves performance by allowing comparison between sparse lexical signals and richer semantic representations, reducing reliance on surface vocabulary and increasing sensitivity to conceptual change.

Axis 2 – Comparison Method

This axis concerns how temporal change is measured. Direct tracking measures change using counts, topic shares, or centroid movement, which is easy to understand but cannot show how features work together. Classification methods identify key differences across time periods but can confuse topic changes and how ideas are presented. Geometric methods (PCA, UMAP) reveal patterns in the embedding space; PCA shows overall differences, while UMAP preserves local groupings, but both primarily describe the data rather than explain it.

Centroid-based drift provides a way to measure change over time that works across different representations, allowing direct comparison between TF-IDF and embedding spaces. This axis makes results more reliable by combining easy-to-understand tracking methods with geometric and predictive approaches, letting us assess time-based changes through both direct measures and hidden patterns.

Sensitivity, Robustness, and the Role of Convergence

Methods differ systematically in sensitivity: lexical approaches capture vocabulary change, embeddings capture semantic shift, topic models capture thematic structure, and classification captures separability but conflates multiple signals. This heterogeneity is used as a design feature rather than a limitation: combining methods reduces dependence on any single representational bias and improves robustness to method-specific noise. As a result, convergence across independent approaches provides stronger evidence of genuine temporal change, while divergence is treated as informative about different layers of discourse evolution.

Four hypotheses structure the evaluation:

H1: Topic models recover the lexical shift from theory to application while revealing transitional hybrid themes.

H2: Embedding-based methods detect conceptual shifts earlier than TF-IDF due to changes in semantic context preceding vocabulary adoption.

H3: Centroid drift captures fine-grained temporal transitions, while classification reflects broader separability rather than precise inflection points.

H4: Major temporal patterns will be broadly consistent across representations and distance metrics, indicating ro-

bustness to methodological choice, while also exhibiting systematic differences in sensitivity.

5 Experimental Evaluation

5.1 Performance Metrics and Their Limitations

The five methods represent fundamentally different analytic paradigms: unsupervised topic discovery, lexical counting, supervised classification, distributional drift, and dimensionality reduction. Each method requires a specific evaluation framework, yet none of the temporal analyses have an external ground truth for verification. Therefore, cross-method convergence is used as the primary standard of evidence. Findings that recur across methods with different representational assumptions carry more evidential weight than those from a single analysis. Table 1 summarises the metric choices and their principal limitations.

Table 2: Performance metrics and key limitations by method

Method	Metric(s)	Key Limitation
M1 Topic	C_v coherence score	Rewards internal consistency; cannot penalise redundant topics or validate manual theory/application labelling
M2 Diffusion	TF-IDF Δ , Word2Vec cosine, SPECTER2 drift	No absolute scale; SPECTER2 coverage constraint excludes most emergent terms by construction
M3 Framing	Macro F1, Precision, Recall (5-fold CV)	Top classifier features reflect broad topical vocabulary, not specifically limitation-framing language
M4 Dim. red.	Pearson r , UMAP topology	Centroid drift collapses within-year heterogeneity; TF-IDF PCA conflates temporal and subfield variance
M5 Lexical	Raw term frequency (normalised by character count); TF-IDF weighted score per category per year	Vocabulary-bound: terms absent from the predefined list are invisible; likely understates the scale of the shift

M1-Topic modelling uses C_v coherence, measuring how often the top words in a topic co-occur in short text segments. C_v matches how people understand short documents and was chosen over held-out perplexity, which favors too many unclear topics and does not work well with short texts. C_v finds the optimal number of topics but cannot evaluate manual labeling as theory- or application-focused; experts must judge that.

M2-Terminology diffusion combines three signals. Changes occur at the surface level through shifts in TF-IDF scores, at the word-meaning level via Word2Vec distances between aligned time periods, and at the overall topic level via SPECTER2 embedding changes. No single measure cap-

tures all three aspects. All scores are normalised to [0,1] across tracked terms; a threshold of (0.5, 0.5) on both axes produces the four-quadrant typology. SPECTER2 (allenai/specter2 via HuggingFace) was selected over general-purpose sentence encoders because its contrastive pretraining on citation pairs directly mirrors the semantic structure of a research abstract corpus.

M3-Limitation framing reports uses macro F1 as the main measure and five-fold stratified cross-validation. Macro averaging matters because the two periods, 2010–2013 and 2018–2020, have very different numbers of abstracts. Accuracy alone would mislead with overly positive results. Precision and recall are reported separately to show types of classification errors.

M4-Dimensionality uses the Pearson correlation between TF-IDF and Sentence-BERT year-to-year centroid distance series to measure consistency between representations. Distances are calculated in full-dimensional space: 50 components for TF-IDF after PCA, and 384 dimensions for SBERT. The 2D TF-IDF projection retains only about 1.6% of total variance, so 2D centroid distances are unreliable. PCA and UMAP are both used for visualization; however, UMAP is only for visual purposes and does not provide numerical results.

M5-Lexical analysis compares raw term frequency (keyword count normalised by total characters per year) and mean TF-IDF score per keyword category. Comparing two scoring schemes is itself part of the experimental design: agreement in direction and timing strengthens the finding, while disagreement would flag a methodological artefact. The fundamental limitation is that the predefined keyword sets cannot capture newer application-oriented vocabulary, such as fine-tuning or zero-shot, almost certainly understating the true scale of the observed shift.

5.2 Experimental Setup

Corpus and temporal coverage. Temporal analyses use the same period structure described in Section 3: M5 and M3 operate on the full available range of 1990–2025 to capture the longest possible trajectory of lexical and limitation-framing change. M1, M2, and M4 operate from 2000 to 2021, where abstract volume is sufficient to support period-level topic models and embedding geometries. Corpus growth is severely skewed, and this is addressed differently across methods. M2 applies reliability weights of 0.27 and 0.49 to the two earliest periods to prevent sparse early corpora from distorting TF-IDF averages and Word2Vec training. M3 uses stratified folds to account for class imbalance. M4 and M5 compute per-year statistics without rebalancing, since the publication growth trajectory is itself a substantive empirical finding rather than a confound to be removed.

Preprocessing. A shared preprocessing pipeline was applied. All methods used the same preprocessing steps described in Section 3, including changing text to lower-case, breaking text into words, removing common words, and using WordNet to reduce words to their base forms based on their part of speech. However, for SentenceBERT

and SPECTER2, the original abstracts, without base-form changes, were retained to preserve sentence structure and meaning. The temporal analyses (M1, M2, M4, M5): temporal ordering is the analytical design, not a holdout partition. M3 is the only exception. It uses a deliberately disjoint comparison of 2010–2013 versus 2018–2020, skipping the 2014–2017 transitional years to suppress noise during the period of change and sharpen the pre/post-neural-turn contrast. Five-fold stratified cross-validation is applied within this split.

Method-specific setup. For M1, the number of topics k was tested with values $\{6, 8, 10, 12\}$ for both LDA (Gensim) and NMF (scikit-learn). The setting that gave the best C_v score was chosen: $k = 10$ for both models. LDA used simple word counts; NMF used TF-IDF features. Topics were labeled by hand, based on the top 10 words for each topic and classifying them as theory-focused, application-focused, or method-focused.

For M2, candidate terms were mined by computing per-period TF-IDF on a concatenated period corpus (`sublinear_tf=True`) and retaining the top-60 rising terms. Approximately 37% were removed as subsumed n -grams, yielding approximately 48 candidates. Before Word2Vec training, 174 multi-word compounds (e.g., `word_embeddings_pre_training`) were merged into underscored tokens to prevent spurious decomposition. Word2Vec skip-gram models (`vector_size=300`, `window=12`, `min_count=2`, `negative=10`, `epochs=30`, `seed=42`) were trained per temporal period; `window=12` was chosen over the default 5 to capture longer-range scientific context, given the longer mean sentence length of abstracts. Cross-period alignment used Procrustes orthogonal transformation (`scipy.linalg.orthogonal_procrustes`) on 1,436 shared anchor terms. For SPECTER2, encoding used raw non-lemmatised abstracts (batch size 64, L2-normalised outputs via `SentenceTransformer("allenai/specter2_base")`), with lemmatisation applied only at the term-detection layer. Bootstrap aggregation resampled abstracts $B = 100$ times per (term, period) cell; cells with fewer than 5 abstracts were excluded, leaving 30 of 48 candidate terms with full SPECTER2 coverage.

5.3 Results

5.3.1 Theory-to-Application Shift

M5, M1, and M4 converge independently on approximately 2014–2016 as the primary inflection point. Under M5, application-oriented keyword prevalence overtakes theory-oriented prevalence in 2014 under raw frequency scoring and in 2015 under TF-IDF, with both measures showing a sustained upward trend through 2025 and no reversal. The one-year difference between scoring schemes falls within the expected noise of TF-IDF’s corpus-relative scaling. The directional finding is stable across both vocabulary sizes tested: a focused 9-term set and a broader 29-term set agree on direction, though the larger set introduced noise from semantically ambiguous terms such as model and method that

appear with nearly equal frequency in theoretical and applied contexts, supporting the choice of the narrower vocabulary. Topic modelling (M1) corroborates this independently and without predefined vocabulary. NMF with 10 topics achieved the highest coherence across all configurations ($C_v = 0.55$, Table 3). Topics dominated by algorithm, optimisation, convergence, statistical, estimation, and inference were labelled theory-oriented; topics dominated by dataset, prediction, healthcare, translation, recommendation, and platform were labelled application-oriented. Tracking proportions over time shows an uninterrupted rise in application-oriented topic share from the early 2000s through 2021. Critically, both LDA and NMF recover this trend despite their different mathematical assumptions. LDA through probabilistic topic assignment, NMF through non-negative matrix factorisation of TF-IDF features, which constitutes evidence that the signal reflects genuine structure in the data rather than the inductive bias of either model.

Table 3: Topic coherence scores by model and configuration

Model	Representation	Topics	C_v
LDA	BoW	6	0.43
LDA	BoW	8	0.46
LDA	BoW	10	0.48
NMF	TF-IDF	6	0.49
NMF	TF-IDF	8	0.52
NMF	TF-IDF	10	0.55

Dimensionality reduction (M4) provides the strongest cross-representation corroboration. The Pearson correlation between TF-IDF and the Sentence-BERT year-on-year centroid distance series is $r = 0.863$, indicating that two fundamentally different representations independently track the same temporal rhythm of change. Both representations place 2014–2016 as the highest-movement window (Table 3). The most counterintuitive result is that 2017 ranks last in both representations despite being the year of the Transformer paper. This is mechanistically principled: the 2017 corpus reflects work already in progress before the Transformer architecture existed. Corpus-based analysis records the diffusion of ideas at scale, not their point of origin. The paradigm shift registers approximately 12 to 24 months after publication, when citing papers adopt the vocabulary at scale.

Table 4: Year-on-year centroid distance rankings (selected years)

Year	TF-IDF	SBERT	Interpretation
2014	0.655 (5th)	0.890 (3rd)	Neural transition begins, SBERT starts showing stronger structure sensitivity
2015	0.754 (4th)	0.935 (2nd)	Peak neural adoption and embedding stabilization effects
2016	0.959 (2nd)	0.831 (4th)	Highest TF-IDF centroid shift, strong lexical drift observed
2017	Lowest	Lowest	Transitional stability period, reduced semantic displacement
2018	High (3rd)	Moderate (5th)	BERT-era emergence with secondary structural realignment

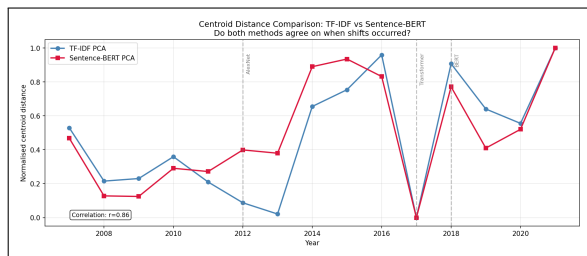


Figure 2: Year-on-year centroid distance comparison, TF-IDF vs. SBERT

While the two representations agree on the timing of major shifts, they differ in sensitivity. Sentence-BERT assigns higher relative movement to earlier years within the transition (e.g., 2014–2015), whereas TF-IDF peaks later (2016), consistent with a lag in vocabulary adoption relative to underlying conceptual change. This supports H4, indicating that semantic and lexical signals are not perfectly synchronous but remain directionally aligned.

Extended Embedding Structure Analysis

Post-hoc analysis of Sentence-BERT embeddings is consistent with TF-IDF PCA and confirming that the largest sources of variance arise from disciplinary differences rather than temporal change. Nearest-neighbour analysis (using the BERT paper as a probe) validates semantic coherence and reveals a diffusion effect, where follow-on work (2019–2020) clusters more closely than contemporaneous papers, indicating that embedding space reflects intellectual influence and reinforcing the finding that corpus-based methods capture the spread of ideas rather than their origin. K-means clustering ($k = 8$) further supports the interpretation that the field has expanded rather than replaced its structure, with declining symbolic AI, stable core task clusters, and emergent neural systems, and safety-related research coexisting over time. Crucially, extending PCA beyond the standard 2D projection shows that temporally coherent signals emerge only in higher-order components, with PC4 exhibiting a strong monotonic trend (Kendall's $\tau = +0.667$, $p < 0.001$) corresponding to the rise of socially-aware AI research (e.g., fairness, interpretability, ethics), a signal entirely invisible

in 2D due to its orthogonality to dominant variance. This demonstrates that meaningful temporal dynamics exist in directions not aligned with the largest variance components, and that low-dimensional projections alone can obscure key patterns of field evolution.

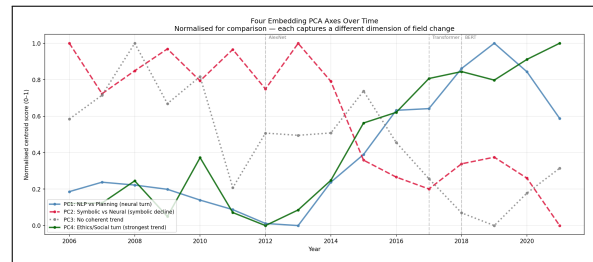


Figure 5: Fig. 4: Embedding PCA components over time

5.3.2 Limitation Framing

The four-category lexical analysis reveals distinct temporal trajectories across framing types. Hedging language (could, may) was consistently present across all periods, suggesting it is a stable feature of academic register rather than a response to any specific development. Capability language (limited, cannot) was the most common category in early periods but declined steadily from approximately 28% in 1990 to 22% by 2020–2021. Failure language (fail, degrade) began low at around 6% and grew to become the second most prevalent category by 2020–2021 at approximately 24%, reflecting the increasing emphasis on empirical benchmarking and performance comparison. Ethics language (bias, fairness) was near-absent before 2015 and rose to $\approx 13\%$ of abstracts by 2020–2021, a trajectory that directly parallels the large-scale deployment of AI systems in socially consequential contexts.

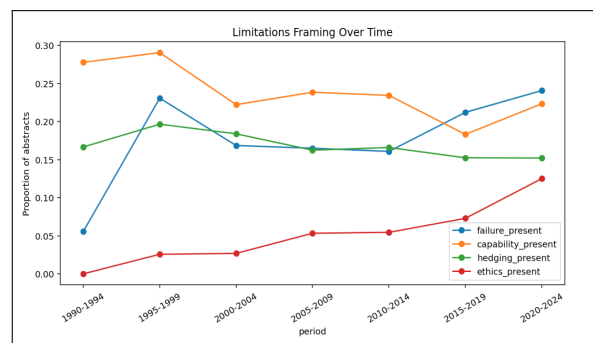


Figure 5: Proportion of abstracts containing each limitation framing category across time. Ethics language shows the sharpest rise post-2015.

Notably, method-introducing and non-method abstracts show near-identical rates of limitation language (51.4% vs. 51.1%), indicating that limitation framing has become a normalised feature of academic writing across research types rather than a marker of methodological modesty in technically focused papers. The binary classifier achieved a mean cross-validation accuracy of 85.8% with macro F1 scores of 0.71 for the early period (2010–2013) and 0.92 for the late period (2018–2020).

Table 5: Classification performance (5-fold CV)

Period	F1	Observation
Early (2010–13)	0.71	Minority class; higher variance; more diverse vocabulary
Late (2018–20)	0.92	The majority class is dominated by consistent neural-era vocabulary
Mean CV Acc.	85.8%	Macro-averaged to correct for class imbalance

The F1 gap reflects the class imbalance between periods and the greater lexical consistency of late-period abstracts, which are dominated by a relatively stable neural-era vocabulary. The high overall accuracy confirms that the two periods are robustly distinguishable from abstract text alone.

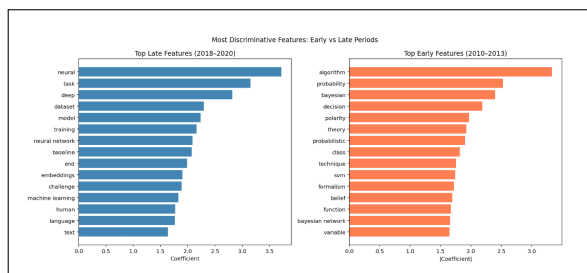


Figure 6: Top discriminative features for the early (2010–13) and late (2018–20) periods. Features reflect broad topical change rather than limitation-specific language.

Cosine distance between five-year period centroids reveals two statistically distinct transitions: 1990–1994 to 1995–1999 (distance = 0.349) and 2010–2014 to 2015–2019 (distance = 0.303). All other consecutive period pairs show substantially smaller distances (0.09–0.19), confirming that the limitation discourse evolves through discrete phase transitions rather than gradual continuous drift. PCA trajectories show stable semantic positioning from 1990 to 2014, a sharp shift from 2015 to 2019, and gradual stabilisation thereafter (distance = 0.090).

5.3.3 Terminology Diffusion

The four-quadrant typology classifies 36 tracked terms across two axes: normalised TF-IDF rise and mean W2V + SPECTER2 drift, thresholded at (0.5, 0.5).



Figure 7: Four-quadrant diffusion typology.

The largest category is Semantic Shift Only ($n = 13$), followed by Stable/Peripheral ($n = 16$), Diffused ($n = 6$), and a singleton Actively Diffusing quadrant occupied by *pre-training*. This distribution challenges frequency-centric accounts of terminology adoption: the dominant diffusion pathway in this corpus is quiet conceptual repositioning that is entirely invisible to frequency-only analysis.

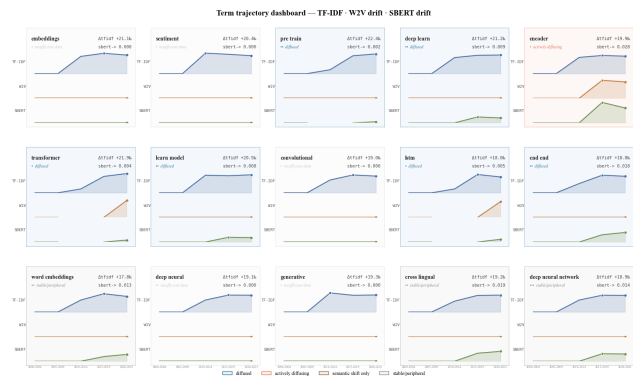


Figure 8: Per-term trajectory cards showing TF-IDF, W2V, and SPECTER2 signals across five periods. Card colour encodes the diffusion quadrant.

Encoder case provides the clearest convergent evidence across all three signals, achieving the highest consensus score (0.63) and highest SPECTER2 drift (0.0424). Visual inspection of the neighbour replacement ribbons confirms near-total neighbourhood replacement across periods Jac-card ≈ 0.001 , demonstrating that the term's distributional context changed completely between the earliest and latest periods.

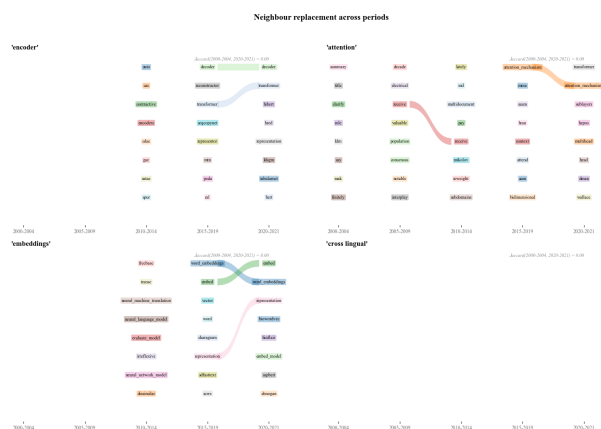


Figure 9: Neighbour replacement ribbons across periods. *Encoder* shows near-total replacement.

The counterexample is pre-training: highest TF-IDF rise (≈ 0.84) but near-zero SPECTER2 drift (0.0021). This identifies it as a discourse template replication event. The BERT fine-tuning argumentative structure was established in a small number of pioneering papers and reproduced at scale with minimal variation, producing a terminological monoculture rather than conceptual diffusion.

5.4 Error Analysis

Topic Modeling Ambiguity (M1)

Manual labeling involves unavoidable personal judgment for mixed topics. A topic focused on neural architecture and deep layers was labeled as methodological but could reasonably fit under either theory or application, since many NLP abstracts report both architectural innovation and practical task use. This is a basic limitation of bag-of-words methods: themes that appear together cannot be separated within a single abstract, so topics that span both sides get a single, approximate label. Theory/application proportion curves should be seen as general trends rather than exact measurements. LDA is more affected by this than NMF, as its lower coherence scores indicate less clear topic boundaries and less consistent labeling across runs.

Diffusion Coverage Bias (M2)

The SPECTER2 rule requires at least 5 abstracts per term-period group excludes new terms such as transformer, downstream task, low resource, and prompt, which did not appear before 2010–14 and thus lack an early reference point for measuring change. This causes a consistent underestimation of the most important terms. The uneven coverage stems from how the metric is designed and cannot be attributed to random error. The Procrustes alignment adds to this problem: only 1,436 of 20,631 combined vocabulary terms (6.9%) appear in all time periods. For terms that change a lot and are surrounded by time-specific words missing from the reference set, alignment errors can still happen unless the size of the reference vocabulary is tested for sensitivity.

Classification Errors (M3)

The confusion matrix shows a clear one-sided imbalance: 108 late-period abstracts were mislabeled as early, while only 35 early abstracts were mislabeled as late. This imbalance

is intentional. The misclassified late abstracts consistently use older vocabulary, such as Bayesian inference, causal modeling, and formal semantics, even in 2018–2020. The classifier learned the main late-period language but misses the smaller group of late papers that avoid neural-era terms, creating a distinct subgroup rather than random errors. Early abstracts are rarely mislabeled as late because their older vocabulary is very distinctive and almost never appears in later work.

A bigger problem appears when looking at the top distinguishing features. The most predictive words are algorithm, Bayesian, probability (early), and neural, deep embeddings (late). These are broad topics, not specific words for limitations. Only one of the top classifier features matches the manually created limitation keyword lists. So, the classifier is tracking the overall vocabulary change in the subfield, not changes in how limitations are described, mixing a related factor with the main signal. This limits how well it can directly show language evolution, even though its overall accuracy is good.

PCA Failure (M4)

Using PCA on TF-IDF vectors reveals a basic limitation. The main components reflect the vocabulary of different subfields instead of changes over time. The first component is influenced by NLP-related words like language, word, translation, and sentence, and separates cs.CL from cs.LG and cs.AI. So, the biggest source of variation shows the mix of disciplines rather than evolution. There is also a significant loss of information: 2D PCA captures only 1.6% of the total variation, and even 50 components capture only about 14%. This reduces the reliability of centroid-based analysis.

Sentence-BERT embeddings show a similar problem. The main components capture differences between disciplines – such as NLP versus broader AI and symbolic versus connectionist approaches not changes over time. Since embedding dimensions cannot be directly understood, they have to be inferred qualitatively. Time-related patterns appear in less important components, confirming that time is not a main source of variation. Variation within each year decreases over time, even as average positions shift, showing the field moves along a common path rather than splitting into separate directions.

5.5 Discussion

Cross-Method Convergence

Importantly, it is not the results of any single experiment that matter, but that all five methods point to 2014–2016 as the key turning point. Word counts (M5) change around 2014–2015; application-focused topic counts (M1) keep rising during this time; TF-IDF and Sentence-BERT centroid distances (M4, $r = 0.863$) reach their highest points independently here; Word2Vec (M2) drift curves show a clear shift from 2010–2014 to 2015–2019 for tracked terms. Despite their different ways of working and scale, their agreement strongly shows that this turning point is a real feature of the data, directly supporting H4.

Expansion, Not Replacement

This is more about growth than replacement. UMAPs show all time periods mixed together in the meaning space. Theory-focused topics make up a smaller part of the field, but still exist; most tracked terms just moved position. At the same time, variation within years decreases, showing that the field is uniting around methods and evaluation practices, adding new application-focused practices while maintaining a core theoretical base. Methods that look for replacement (like losing keywords) will overestimate how much the field is breaking apart.

Hypothesis Results

H1 verified: the NMF recovered the paradigm shift and hybrid topics that were missed by the keywords. H2 confirmed on the encoder case, but SPECTER2's coverage constraint blocks testing the most emergent terms. H3 partially confirmed: centroid measures captured the 2017 lag, but classification captured between-period separability but conflated topical and framing effects. H4 is very strongly supported by five-fold convergence on 2014–2016.

Comparative Strengths and Recommendations

Manual vocabulary analysis (M5) is easy to understand but depends on specific words, so it gives a minimum estimate of the shift. Topic modeling (M1) removes the vocabulary limit, and agreement between LDA and NMF builds confidence, but the bag-of-words model mixes topics that appear together, and naming them is subjective. Dimensionality reduction (M4) is the best method for cross-checking results, but it should be checked to ensure changes are not due solely to shifts in subfields. Terminology diffusion (M2) is the most unique: the 13-term Semantic Shift Only group and the pre-training template replication result appear only in diffusion; the limited Procrustes anchoring of diffusion (6.9% of the combined vocabulary) cannot rule out alignment errors for terms with large changes. Classification (M3) is the only method with a numeric measure of how well it generalizes, but it mixes topic words with language specific to limitations.

Three improvements would greatly help future work: expanding the collection beyond 2021 to clarify ongoing diffusion trends for cross-language and benchmark; replacing simple two-period classification with a sequence model to find when framing changed instead of just showing two separate periods; and repeating analyses on full paper texts to see if the almost identical amount of limitation language in different paper types (51.4% vs. 51.1%) means normalization or abstract-genre normalization.

6 Conclusion

This study examined how discussions of AI, ML, and NLP research changed over 30 years by analyzing 24,677 arXiv abstracts using five text analysis methods. These methods together provided a clear and consistent perspective, providing the basis for the findings discussed below.

The clearest finding is that 2014 to 2016 represented a real turning point, shown independently by word analysis, topic grouping, classification, tracking term spread, and data simplification methods. Notably, a strong link of 0.863 between

TF-IDF and Sentence-BERT distance measures, which work at very different language levels, is the best proof that this change is a real part of the field's structure, not just a result of the methods used. This change is better described as an increase rather than a replacement: theory-focused topics became less common but remained present, and most tracked terms shifted in meaning rather than disappearing. Building on these results,

The analysis of how terms spread gave the most new insight. Specifically, combining changes in TF-IDF, shifts in Word2Vec word groups, and changes in SPECTER2 discourse showed different ways terms were adopted that no single method could see. The main pattern, quiet shifts in meaning without matching increases in use, explained 13 of 36 tracked terms. The encoder example showed this clearly, with a 63-fold increase in mentions and a complete change in its role, visible only through SPECTER2. The pre-training example showed that copying templates is a different kind of change, which a simple frequency count would have missed. Looking at how limitations were described added another view: ethics-related words grew from almost none before 2015 to 13% of abstracts by 2020–2024, and failure-related words became the second most common way to describe limitations, changes not visible in simple counts of limitation words. All methods performed as intended. The limitation of the framing classifier was that it measured broad subfield vocabulary shifts rather than framing-specific change, and TF-IDF PCA recovered subfield composition variance rather than temporal drift. Both failures were diagnostically informative rather than simply negative, demonstrating the value of cross-method checking. Turning to the remaining challenges,

Three difficulties remain. The SPECTER2 coverage limits exclude the newest terms, such as transformer, prompt, and downstream task, by design. The 6.9% Procrustes anchor basis can make some findings about term changes unclear because of alignment issues. Looking only at abstracts may hide the full variety of how limitations are described in full papers. The bigger lesson is that no single method fully captures how scientific language changes. The strength of this study lies in the concurrence across methods, and their disagreements were also helpful.

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